Embodied AI Must Treat Visual Bias as a Safety Primitive, Not a Robustness Penalty

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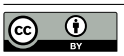
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Abstract

While embodied agents increasingly incorporate diverse sensors, they predominantly rely on learned visual representations for manipulation and control. However, when visual preferences are tightly coupled with action generation, benign cues can induce task failures, unsafe motions, or exploitable vulnerabilities. We present a diagnostic study of visual bias as a *safety primitive* for embodied AI evaluation and defense: an independently operationalizable property tied to an identifiable risk class, not a standalone safety guarantee. Agents should be assessed not merely on task success, but also on whether their actions remain grounded in visual evidence relevant to the task. We study this risk through controlled visual variations under task-equivalent conditions, naturalistic distractors aligned with inherited visual preferences, and counterfactual action analysis. Our results show that benign perceptual changes can produce divergent behavior, while task-irrelevant content can receive disproportionate action relevance and induce physical failures. We further introduce a counterfactual detection method that combines persistent region selection with changes in predicted actions after masking, helping distinguish behaviorally relevant preference-aligned content from regions that are merely visually salient. These findings provide preliminary diagnostic evidence of visual bias and demonstrate the need for broader preference discovery, visual shortcut mining, and failure mode search to improve predeployment risk coverage and guide targeted mitigation in both model development and real-world deployment.

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CCS Concepts

• **Computing methodologies** → *Intelligent agents*; • **Computer systems organization** → *Robotic autonomy*; • **Software and its engineering** → *Software safety*.

Keywords

embodied AI safety, visual bias, vision-language-action models, visual grounding

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1 Introduction

Embodied AI aims to build agents that perceive, reason, plan, and act through interaction with physical or simulated environments [7, 40]. Recent safety discussions emphasize that embodied failures often arise at the system level: perception, decision making, and execution are coupled in ways that allow small errors to propagate into physical behavior [23, 29]. These agents may use language, proprioception, tactile sensing, depth, and other sensory streams, while visual perception often remains a primary interface through which they identify objects, spatial relations, affordances, obstacles, and hazards [15, 46, 54]. This makes visual evidence a central safety concern. A flawed visual judgment is no longer only an incorrect prediction. Once it enters a closed-loop system, it can shape motion, contact, navigation, and interaction with humans or objects.

The coupling between perception and action also explains why the recent optimism around scaling should be treated with care. Across vision-driven embodied domains, including autonomous

driving, visual navigation, robotic manipulation, and generalist robot learning, recent progress is increasingly shaped by large pretrained foundations. Two patterns are especially influential. In one, vision language models (VLMs) serve as planners, reasoners, or grounding modules that organize downstream perception and control. In the other, policies trained end-to-end map visual observations and language instructions more directly into actions, with vision-language-action (VLA) models such as RT-2 [60] and OpenVLA [19] as prominent examples. Scaling data, model capacity, and cross-embodiment pretraining has clearly expanded capability and remains a promising direction for embodied AI [32, 38]. Our claim is not that scaling and auditing are mutually exclusive: broader pretraining and augmentation may improve robustness, while targeted audits test whether the remaining visual evidence dependence is acceptable. Capability scaling alone, however, is not a safety argument. Higher average performance does not guarantee reliable visual evidence for action, and larger pretrained models may still carry biases learned from internet-scale multimodal data into embodied control.

The visual source of this risk precedes embodiment. Static vision models and VLMs are known to rely on non-causal visual regularities, including texture, background context, co-occurrence patterns, visible attributes, and other cues that may correlate with training labels or answers without being physically relevant to the task [11, 12, 50]. Controlled VLM studies identify shortcut features and region-level spurious correlations, while SpuriVerse shows that strong large vision-language models remain vulnerable to previously unseen spurious patterns [5, 31, 44, 49]. Work on language bias likewise asks whether multimodal models actually use the visual evidence needed for an answer [57]. In passive recognition or question answering, such defects may appear as a wrong label, biased answer, or ungrounded rationale. Embodied AI changes the consequence of the same defect. Once visual foundations support planning, grounding, or action generation, a visual preference that was previously confined to prediction space can influence a grasp, a steering command, a route choice, or a tool interaction.

This paper focuses on visual bias as an internal misallocation of visual evidence under otherwise valid observations, rather than a generic failure under external disturbance. Robustness usually asks whether a system can tolerate degraded sensing, adversarial perturbations, or distribution shifts. Visual bias asks a narrower causal question: when task relevant visual evidence remains visible, feasible, and semantically unchanged, does the policy assign it appropriate behavioral weight, or can task irrelevant but visually preferred content alter the action? Related warning signs already appear across embodied domains. In autonomous driving, semantically equivalent viewpoint shifts can degrade end to end trajectory planning despite unchanged road layout [14], while shadows, reflections, and tire marks can mislead lane perception and downstream driving behavior [55]. In visual navigation, changes in camera pose or target appearance can reduce performance even when the same navigation goal remains available [35, 42]. These cases share a failure to keep action grounded in the right visual evidence once pretrained perception is connected to control.

To study this mechanism with minimal ambiguity, we use robotic manipulation as a diagnostic setting. Manipulation is not the boundary of embodied AI, but it provides a controlled setting in which visual evidence is quickly translated into contact, grasping, placement, and object interaction. This makes it useful as a high resolution slice of a broader safety problem. Our study examines two widely used embodied paradigms: systems in which VLMs act as high level planners or grounding modules before downstream execution, and end to end policies built on visual and language pretrained foundations. These paradigms are not confined to tabletop manipulation; they increasingly shape navigation, autonomous driving, aerial robotics, and generalist robot learning. Within this controlled setting, we show that inherited visual preferences can destabilize task equivalent execution and alter generated actions. This evidence points to a mechanism that is not local to one benchmark, but rooted in how pretrained visual foundations are coupled to physical control. These findings motivate treating **visual bias not as a residual robustness penalty to be averaged away, but as a safety primitive for assessing whether physical action is grounded in appropriate visual evidence.** This shifts evaluation from whether an agent completes a task to why it acts as it does and whether its visual basis remains reliable before deployment.

Paper outline. We first distinguish visual bias from generic robustness, arguing that the key safety question is not only whether an agent withstands external disturbance, but whether its actions remain grounded in the right visual evidence (Section 2). We then show why success based evaluation can hide this failure mode, since task completion may obscure unstable visual grounding and unsafe behavior before the final outcome is recorded (Section 3). Building on this distinction, we use robotic manipulation as a diagnostic setting to trace how visual bias enters embodied control (Section 4). Our evidence shows that average success can hide visually driven control instability, and that task irrelevant visual preferences can influence generated actions. We next discuss why scaling alone leaves the visual safety margin undefined (Section 5), and outline directions toward visual causal decoupling through rationale audits, adaptive sensing, and verifiable evidence certificates (Section 6). We close by discussing why broad robustness evaluation does not eliminate the need for separate visual-bias auditing, and why this concern remains relevant as foundation policies are reused across embodied domains (Section 7).

2 Why Visual Bias Cannot Be Treated as Generic Robustness

2.1 The Umbrella of Generic Robustness

Deploying embodied agents in unconstrained physical environments demands strong generic robustness. This broad operational requirement encompasses a system's capacity to maintain reliable performance under distribution shifts, environmental variation, unexpected physical conditions, and degraded or adversarial observations. In practice, this means policies must successfully navigate heavy physical clutter [9, 34], adapt to disruptive weather phenomena [30], and tolerate substantial variations in camera positioning [14]. It also includes maintaining stable behavior when changes in visual appearance expose reliance on spurious or non-causal features. From a technical standpoint, behavioral failures

triggered by visual bias therefore remain part of the broader robustness taxonomy. Our analysis does not remove visual bias from this umbrella. Instead, it argues that this subset should be separately diagnosed and reported when aggregate robustness scores would otherwise conflate failures caused by inappropriate visual evidence allocation with those caused by genuinely degraded, shifted, or infeasible observations. Generic robustness measures overall resilience across operating conditions, whereas visual-bias evaluation isolates whether actions remain grounded in task-relevant visual evidence under task-equivalent changes.

2.2 Visual Bias as Misallocated Visual Evidence

Subsuming visual bias under generic robustness masks a practical distinction. Generic robustness mainly concerns resilience to external uncertainty, such as clutter, sensor degradation, or environmental disturbance. Visual bias, by contrast, is a misallocation of visual evidence: the task-relevant physical information may remain visible and unchanged, yet the policy assigns behavioral weight to incidental appearance. A target remains physically reachable when its surface color changes, and a decorative poster remains irrelevant to a manipulation instruction. Similarly, failure under semantically equivalent camera viewpoints is not necessarily caused by unpredictable occlusion; it may instead reflect weak grounding in three-dimensional structure and an overreliance on shortcuts learned from two-dimensional appearance [42]. Visual bias therefore asks a narrower diagnostic question than generic robustness: are task-equivalent visual alternatives treated comparably, and can task-irrelevant visual content influence action generation? This makes the distinction operational rather than merely taxonomic. It matters because it changes the required intervention. Scaling and data augmentation can broaden exposure to nuisance variation, but they do not guarantee that a policy will rank task-relevant physical structure above spurious visual salience. Since embodied agents often build on shared pretrained visual foundations, action systems can inherit internet-scale style, texture, and saliency regularities [22]. Without structural constraints, broader data can preserve or even amplify these priors rather than remove them. Related work on low-data VLA post-training offers a complementary warning: standard fine-tuning can over-specialize to narrow demonstrations and induce concept or spatial lock-in, causing policies to follow trained objects, attributes, or locations rather than the current instruction [16]. Thus, data and adaptation alone do not automatically eliminate visual bias; they should be paired with mechanisms that preserve and audit visual grounding.

Here, *safety primitive* denotes a distinct dimension of safety assessment requiring separate measurement and reporting, rather than a mechanism with specified guarantees. The question is not only whether an agent withstands disturbance, but whether its actions are grounded in task-relevant evidence. The proposed primitive complements rather than replaces broad robustness: a system may need both high overall resilience and evidence that this particular source of visual instability has been identified and managed.

3 The Success-Centered Illusion

Success rate is an outcome statistic: it tells us how often a benchmark goal is reached under a given protocol. What it leaves open is

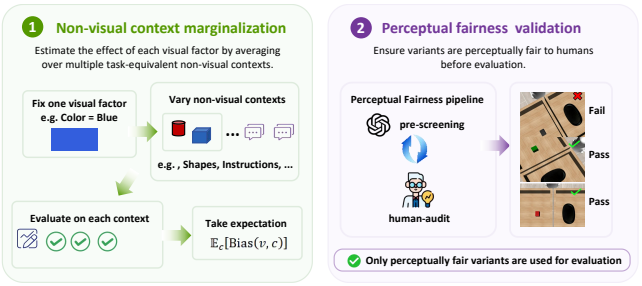


Figure 1: Overview of the visual bias diagnostic probe.

how those successes were produced, and what happened before the final state. This distinction matters when a task score is used to support a broader safety claim. As recent critiques of attack success rate argue, an observed rate can describe results under a test protocol, but it does not automatically measure the safety concept one wants to evaluate [8, 45]. In embodied AI, this gap is especially important because visual evidence is converted into physical motion.

- **The route to success remains unmeasured.** The same success label can arise from very different internal routes. One policy may act from task geometry and object relations, while another may depend on color, texture, viewpoint, background, or other incidental cues. This is why strong benchmark performance can still coexist with sensitivity to clutter, lighting, viewpoint, object appearance, and related visual changes. For visual bias, the missing question is not only whether the goal is reached, but whether task-irrelevant appearance changes the behavior.
- **Completion does not guarantee safe execution.** A successful rollout can still contain overreach, near collision, wrong contact, hesitation, or other unsafe intermediate behavior. Recent embodied safety benchmarks therefore move beyond terminal success by evaluating dynamic risks, unsafe process states, and trajectory-level safety [28, 37, 51]. For visual bias, this matters because a visually driven error can create physical risk before it appears as a task failure.

This diagnostic target differs from, but complements, two established practices. Outcome-oriented safety constrains known hazardous behavior through rules, safety costs, runtime monitors, or trajectory checks; it asks whether actions violate a specified safety boundary. Robustness evaluation varies appearance, viewpoint, layout, or sensing conditions and asks whether performance is preserved. Neither question necessarily identifies whether a policy systematically depends on a particular incidental visual cue. A perturbation may also occlude or remove information, making the task genuinely harder, while aggregate success rates can mask opposing condition-specific effects. Our experiments therefore retain task meaning, target visibility, and physical feasibility, then ask whether unnecessary appearance receives behavioral weight.

4 Diagnostic Evidence: How Visual Bias Enters Embodied Control

Figure 1 summarizes our audit design: we hold task meaning and physical feasibility fixed while varying only visual appearance. This

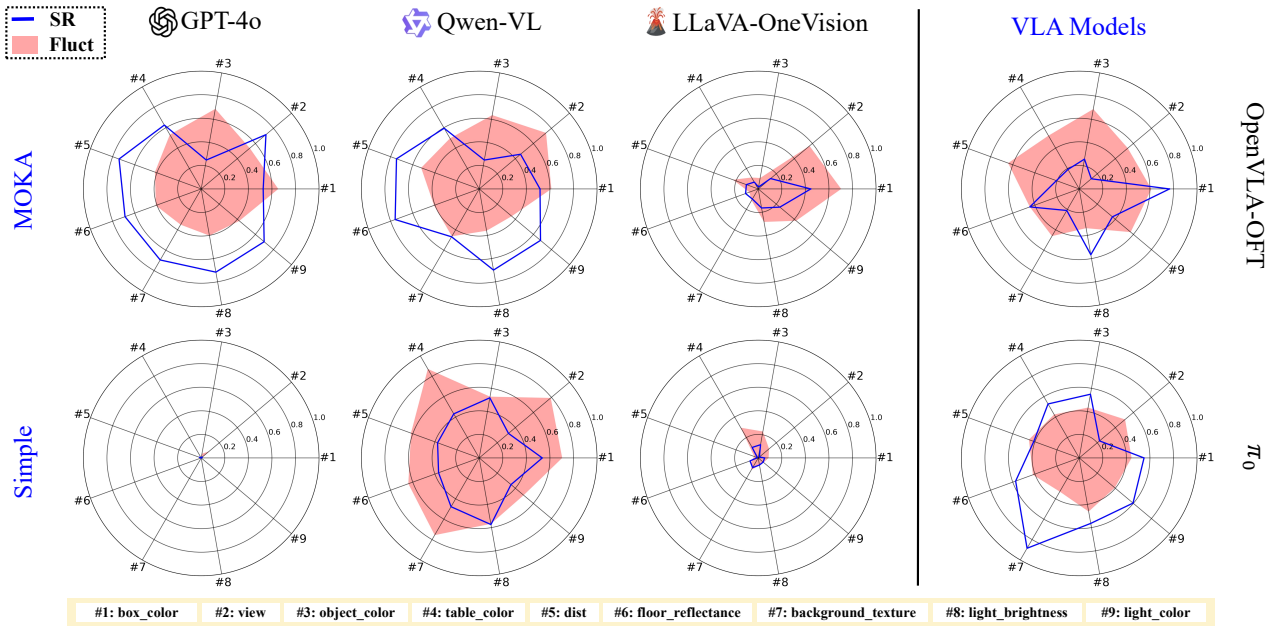


Figure 2: Average task success and visual sensitivity across evaluated embodied agents. Each radar plot reports the normalized success rate (SR, blue) and success variation measured as 2σ (Fluct, red) over nine controlled visual factors.

design isolates appearance as the main source of behavioral change: the agent receives the same instruction, interacts with the same type of target, and faces a task that remains physically executable. If benign appearance shifts still alter execution under these controlled conditions, the failure cannot be explained by a changed goal, an infeasible scene, or a harder manipulation problem. It instead indicates that incidental visual features are affecting how the agent grounds the task and selects actions. In this setting, visual bias becomes a concrete, separately diagnosable control defect within the broader robustness problem.

4.1 Robotic Manipulation as a Diagnostic Microscope

Robotic manipulation provides a sharp test because visually induced errors have little room to dissipate before affecting execution. In driving and navigation, visual brittleness under viewpoint, weather, or appearance variation is relevant to safety, but its effect can be partially absorbed by later observations, spatial margins, route correction, or exploration dynamics. In open-world robotics, clutter and salient distractors similarly expose fragility, but the failure often mixes visual preference with scene complexity [26]. Manipulation reduces this ambiguity. Small changes in visual grounding can appear quickly as errors in contact timing, grasp alignment, placement precision, or object interaction. Therefore, when instruction, target role, and physical feasibility remain unchanged, behavioral changes under benign appearance variation directly signal that incidental visual features affect embodied execution.

A Controlled Visual Bias Diagnostic Probe. We instantiate the diagnostic microscope with a controlled probe that tests whether

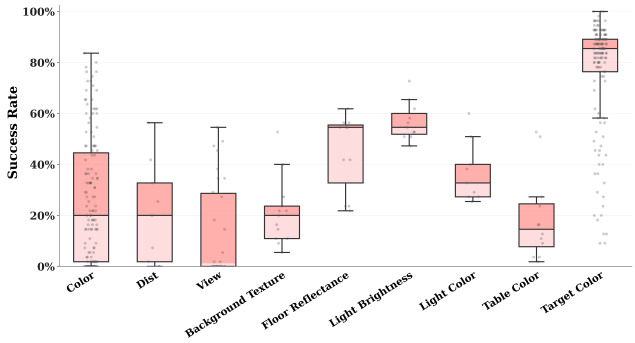


Figure 3: OpenVLA-OFT success-rate distributions across nine task-equivalent visual factors.

appearance alone can alter embodied behavior while task semantics and physical feasibility remain fixed. We evaluate both **VLM-driven agents**, including MOKA [25] and a minimal embodied prototype adapted from the BadRobot [52] setting, which we refer to as **Simple Agent**, and **VLA policies**, including OpenVLA-OFT [18] and π_0 [4]. This allows us to examine whether inherited visual preferences affect both explicit grounding pipelines and direct action generation under the same controlled setting.

Instead of random perturbations, we vary interpretable visual factors that are relevant to embodied fragility. Inspired by STAR-Gen [10], the probe covers nine factors, including object color, camera viewpoint, table color, and lighting, drawing on perturbation dimensions studied in THE COLOSSEUM [34] and LIBERO-Plus [9]. The goal is not to create a generic stress test, but to examine whether task-equivalent visual alternatives are treated consistently. Each

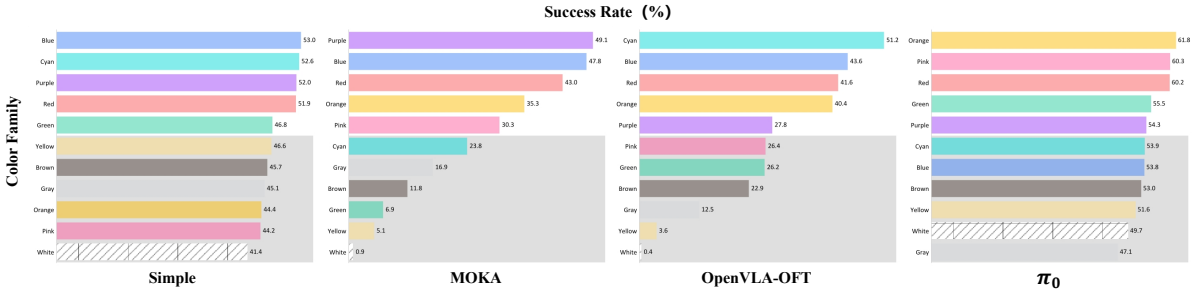


Figure 4: Success rates across target colors and agents. Gray and white targets tend to produce lower success rates, indicating uneven sensitivity to target color.

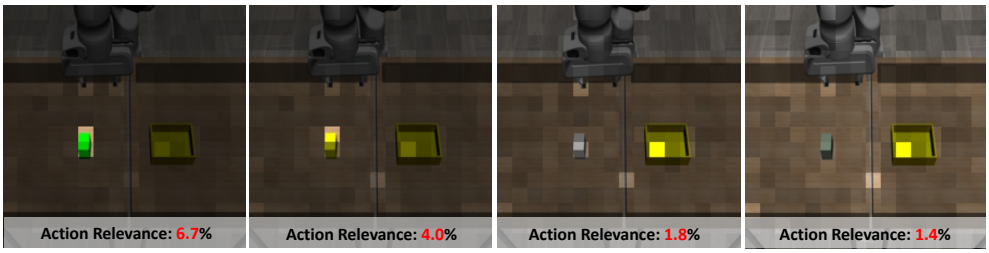


Figure 5: Action relevance evidence in the representative OpenVLA-OFT setting. Under identical task configuration, task-equivalent targets receive unequal internal weighting.

factor is evaluated across non-visual contexts such as target position and instruction wording. To reduce perceptual and physical confounds, candidate variants pass a two-stage fairness protocol before evaluation. An automated multimodal screen checks that the target remains visible and recognizable, the instruction remains feasible, essential content is not occluded, and object identity is unambiguous. Human reviewers then verify task equivalence, target visibility, physical feasibility, and the absence of camouflage-like conditions. We refine the generation configuration whenever either stage rejects more than 5% of candidates, and proceed only after both stages independently exceed a 95% pass rate. Rejected variants include cases in which target and background become too similar, viewpoint or lighting removes critical evidence, or scene changes alter the manipulation geometry. Consequently, the admitted variants preserve task semantics, target identity, perceptual visibility, and physical feasibility as closely as the protocol can establish.

We report the resulting instability as a two-sigma success fluctuation rather than introducing a new capability metric. Let $SR_\pi(v, c)$ denote the success rate of policy π under visual value v and non-visual context c. For a visual factor V_i, we compute $\mathrm{Fluct}_\pi^{2\sigma}(V_i) = 2\,\mathbb{E}_{c\in C_{\mathrm{Gen}}(V_i)}[\mathrm{Std}_{v\in V_i}(SR_\pi(v, c))]$, i.e., twice the expected standard deviation of success across task-equivalent visual alternatives after marginalizing over non-visual contexts. A small fluctuation indicates stable behavior under benign visual changes, whereas a large fluctuation indicates that incidental appearance has become behaviorally consequential. This summary complements average success rate; Figure 3 shows the underlying OpenVLA-OFT distributions.

Probing Action Interference with Naturalistic Distractors. We next examine whether task-irrelevant but visually preferred content can interfere with action generation. Stable Diffusion 3.5 [39] generates 20 prompt variants per semantic target across morphology, material, style, texture, and saturation, and Segment Anything [20] extracts the foreground object. We resize the resulting patch to occupy about 10% of the observation and rank candidates by action-generation relevance using the transformer attribution method of Chefer et al. [6]. Preference-aligned seeds are screened in the representative OpenVLA-OFT+LIBERO-10 setting and then transferred without category re-selection or adversarial refinement to OpenVLA [19] and OpenVLA-OFT across all four LIBERO [24] suites. For every task, the insertion location is manually annotated so that the patch does not cover task objects, occlude essential evidence, or alter the physical scene. The random-noise control uses the same footprint and location, allowing the comparison to distinguish semantic preference from generic image corruption.

Counterfactual Patch Detection through Action Difference. We further construct a counterfactual detection task to distinguish patched queries from clean queries and examine whether visually preferred content affects predicted actions. For each observation and instruction query, a three gate procedure first uses PAD localization [17] to propose candidate regions, retains attention that persists under both the task instruction and an unrelated counterfactual instruction, and combines the localization and persistent attention scores into a selected mask. This procedure identifies persistent visual attractors whose salience is not determined solely by the task instruction.

We replace the selected pixels with a uniform black mask and compare the original and masked action token distributions over a rolling window of $W = 8$ inference steps. The resulting Action-Diff

Table 1: Comparison of failure rates (%) under clean, random noise patch, and benign preference aligned naturalistic patch conditions.

	OpenVLA					OpenVLA-OFT				
Condition	Spatial	Object	Goal	LIBERO-10	**Avg**	Spatial	Object	Goal	LIBERO-10	**Avg**
Clean	15	12	21	46	23.5	2	1	2	4	2.3
Random Noise Patch	29	15	21	48	28.3	27	0	10	14	12.8
Naturalistic Patch	69	89	56	76	**72.5**	24	0	13	38	**18.8**

score, $s_t^{\text{query}} = s_t^{\text{dist}} + s_t^{\text{collapse}}$, combines Jensen–Shannon divergence, action-token argmax flip rates, expected decoded action shift, and entropy change with a lower tail collapse term for abnormally low translation dynamics such as freezing.

Patched queries are treated as positives and clean queries as negatives. For each LIBERO suite, we set the Action-Diff query threshold on clean queries at a 10% false positive rate. At the episode level, above threshold queries are accumulated, and an episode is flagged when their count reaches $K \in \{5, 10, 20, 30\}$. Rollouts are capped at 220, 280, 300, and 520 steps for LIBERO-Spatial, LIBERO-Object, LIBERO-Goal, and LIBERO-10, respectively. We compare Action-Diff with Region-Trigger, which relies only on persistent region selection. Improved discrimination from Action-Diff would indicate that preference aligned content is not merely visually salient, but also affects action generation.

4.2 Evidence I: High Success Masks Unstable Visual Control

Average success can hide physically unacceptable instability. For embodied agents, a policy that succeeds on average but changes behavior sharply under benign visual alternatives is not merely statistically unstable; it is unsafe for physical deployment. Figure 2 illustrates this gap. On standard LIBERO, OpenVLA-OFT and π_0 both achieve success rates above 95%. Under our task-equivalent visual variation, however, OpenVLA-OFT exhibits larger two-sigma fluctuations than π_0 on most of the nine probed factors. Figure 3 resolves this summary into OpenVLA-OFT's factor-wise distributions: several factors have broad interquartile ranges, and the color, distance, and viewpoint probes include near-zero outcomes even though other variants remain much stronger. Thus, high aggregate capability can coexist with sharp condition-specific degradation while the instruction, target role, and manipulation geometry remain unchanged. In physical interaction, such instability is unacceptable because incidental appearance can translate into unpredictable contact, trajectory, or grasping failures.

Embodied agents carry inherited visual preferences into control. Figure 4 shows that the instability is not random dispersion, but a structured preference pattern shared across embodied regimes. Achromatic or weakly saturated target colors, such as gray and white, tend to produce lower success than more visually salient colors. For VLA policies, this preference can enter control directly because visual observations are mapped into action generation. For VLM-driven agents such as MOKA, the same inherited preference appears through a different route: it first affects task decomposition, grounding, target selection, or intermediate visual reasoning, and then propagates into downstream execution. In our MOKA case inspection, the high-level planner more consistently names common colors but more often falls back to generic descriptions such as "geometric object" or "block" for other visually equivalent targets. Holding the image fixed, we then replace the generated color description with neighboring hues and re-query Grounding DINO [27]; a substituted hue exceeds the original grounding confidence by more than 0.03 in 17.78% of cases. This provides a concrete route from uneven description to deviated target localization. Figure 5 provides internal evidence for the VLA case. The resulting action-relevance estimates show that OpenVLA-OFT assigns unequal internal weight to functionally equivalent colored targets. The weaker behavioral conditions also receive weaker relevance: green receives 6.7%, yellow 4.0%, white 1.8%, and gray only 1.4%. Together with the VLM-driven case analysis, this alignment indicates that embodied agents are not merely encountering color-specific difficulty; they inherit visual preferences from pretrained visual foundations and carry those preferences into physical control.

4.3 Evidence II: Distractors Aligned with Visual Preferences Expose a Physical Attack Surface

The distractor results show how inherited visual preference can turn into physical risk. Although each inserted patch occupies only about 10% of the image and is unrelated to the task, some naturalistic patches receive more than 20% of the estimated action-related relevance. When these high-relevance patches are inserted into LIBERO observations, Table 1 shows a large increase in failure rate without adversarial optimization. On OpenVLA, the average failure rate rises from 28.3% under random noise patches to 72.5% under preference-aligned naturalistic patches. OpenVLA-OFT is more resistant, but its failure rate still increases from 12.8% to 18.8%. The contrast with random noise indicates that the effect is not caused by generic visual corruption. It appears when task irrelevant content matches the model's visual preference and competes with task evidence. Figure 6 further shows that this degradation corresponds to concrete physical failures, including dropped objects, tipped containers, repeated manipulation failure, and unsafe arm motion.

This result shows why visual preference can become exploitable. Evidence I showed that task equivalent visual alternatives are weighted unequally; here, an irrelevant pattern that falls on the preferred side of that weighting can already degrade physical execution. Existing physical attack studies have shown that embodied systems can be manipulated by external visual artifacts [13, 56]. Our finding highlights a complementary risk: the vulnerability can begin from ordinary visual content that the policy already treats as important, before any explicit optimization is applied.

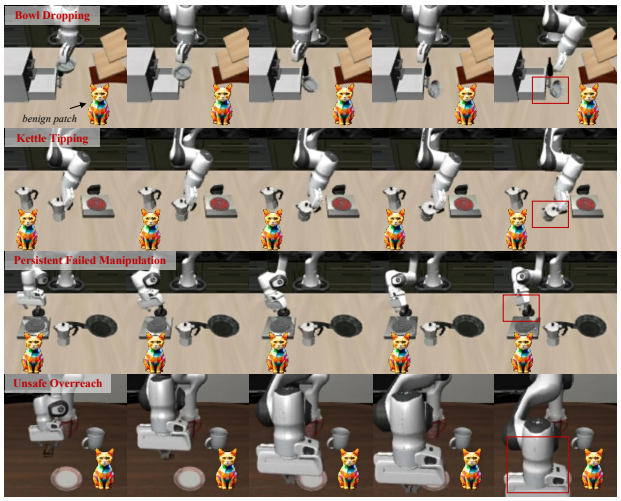


Figure 6: Representative unsafe physical outcomes under benign preference aligned patch conditions.

4.4 Evidence III: Counterfactual Detection Links Visual Preference to Action Generation

We evaluate whether counterfactual action differences can distinguish preference-aligned patches from visually salient but behaviorally irrelevant content. The query-level evaluation contains 650 patched queries as positives and 378 clean queries as negatives. Figure 7 shows that the Action-Diff score separates the two groups with an AUROC of `0.967` on LIBERO-10, `0.980` on LIBERO-Object, and `0.962` on LIBERO-Spatial. LIBERO-Goal is more challenging, with an AUROC of `0.778`, indicating that its goal-conditioned tasks produce less separable action responses. Each star reports the TPR obtained when the threshold is fixed at a `10%` false-positive rate on clean queries. Figure 8 further compares Action-Diff with Region-Trigger at the episode level. Region-Trigger accumulates persistent suspicious-region selections without measuring whether those regions affect the predicted action. It can therefore respond to visually attractive regions that also appear in clean observations. Action-Diff instead incorporates the change between the original and masked action distributions. Component ablations in Appendix A further support this design: removing terms that capture distributional change or collapse increases false positives or weakens detection, indicating that these terms help distinguish action-relevant preference from mere visual salience. As the required anomalous-query count K increases, Action-Diff substantially reduces false-positive episodes while retaining high true-positive counts in most suites.

5 What Goes Wrong If Scaling Remains the Answer

Unknown visual safety margins. If scaling remains the default answer, visual bias becomes a hidden safety debt rather than a solved problem. Larger models and broader data may improve average success, but they do not specify when an agent's visual evidence becomes unreliable. This matters because embodied errors do not stay in prediction space. In autonomous driving, biased visual evidence can affect braking, lane keeping, or obstacle avoidance under unchanged road semantics [14]. In visual navigation, misleading surfaces or landmarks can redirect an agent despite unchanged spatial goals [42]. In robotic manipulation, the same defect reaches grasping, contact, tool use, and human-facing interaction, where visually biased actions directly affect the physical world [21, 22, 56]. The issue is therefore not that scaling is useless, but that it can leave the boundary of safe visual grounding undefined. Scaling and targeted auditing should be complementary: scaling seeks broader capability and invariance, while auditing measures whether identifiable visual-evidence failures remain. Without task-equivalent visual tests, deployment inherits an unknown safety margin.

Wrong interventions after physical failure. When visual bias is not diagnosed directly, physical failures are easy to explain away as long-tail robustness. A robot that drops an object, tips a container, collides with nearby structure, or overreaches into unsafe space may be attributed to clutter, unusual viewpoint, or external interference. This pushes the field toward more data, broader augmentation, or stronger stress testing, while leaving the evidence ranking behind action unexamined. As a result, the same failure mechanism can then reappear whenever task-irrelevant visual content aligns with inherited model preferences, leaving a latent physical instability that scaling alone may not remove.

6 Path Forward: Toward Visual Causal Decoupling

Rationale audits for embodied action. Future evaluation should move beyond aggregate outcomes and audit the visual evidence supporting action. A policy should not be considered safe merely because it reaches the goal; it should also preserve behavior under task-equivalent visual alternatives and avoid relying on task-irrelevant visual cues. Controlled probes are only a first step. The broader requirement is to evaluate whether success is supported by causally appropriate visual evidence.

Adaptive sensing for reliable grounding. Adaptive sensing offers a practical way to reduce visual bias before it enters control. Recent work argues that robustness can be improved not only by scaling models and data, but also by actively shaping the observation process through camera choices, sensing configurations, and perception feedback [1–3]. For embodied agents, unstable visual evidence should prompt the system to gather better observations rather than act immediately. By changing viewpoint or incorporating complementary sensing before execution, the policy can base its action on more reliable task evidence and reduce the influence of incidental appearance on physical behavior.

Verifiable visual resilience. Within this direction, ZKML [33] and zkLLM [41] can turn visual evidence checks into verifiable certificates. Their value is not in proving semantic grounding directly, but in verifying faithful execution of specified inference and attention-related computations. For embodied agents, this enables a concrete certificate interface: before action release, the system can verify that the proposed action, task description, estimated object regions, scene geometry, and policy evidence ranking satisfy a predefined grounding predicate. Such a predicate could require

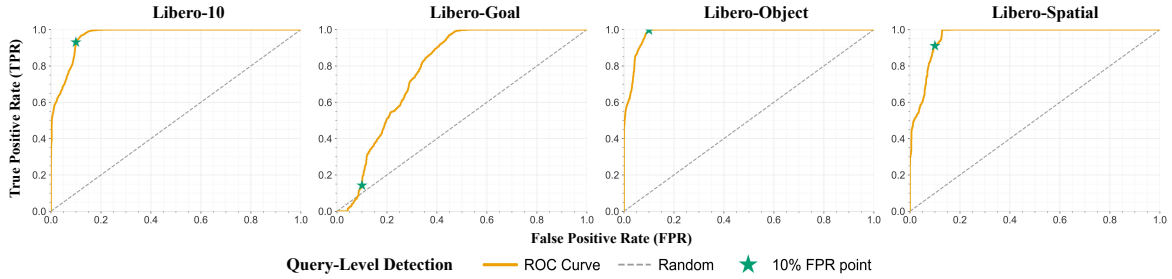


Figure 7: Query-level counterfactual detection on OpenVLA. Patched queries are positives and clean queries are negatives. Each suite uses its own threshold at 10% clean-query FPR, and the star reports the corresponding TPR.

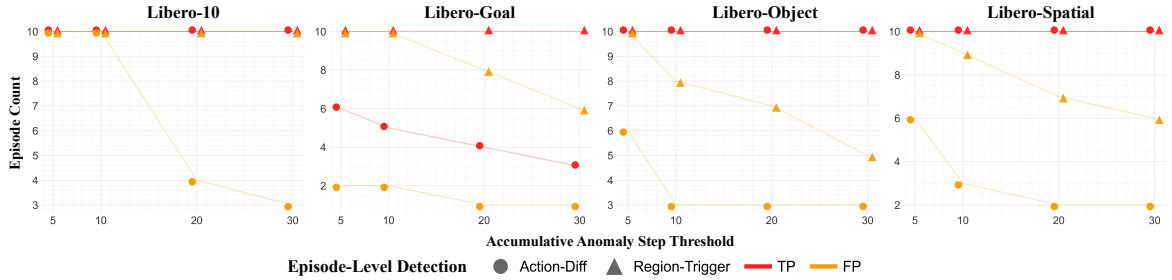


Figure 8: Episode-level detection using each suite's 10% clean-query-FPR Action-Diff threshold. The horizontal axis gives the accumulated anomalous-query count K required to flag an episode. Region-Trigger uses persistent region selection alone, whereas Action-Diff also incorporates the action response after masking.

dominant evidence on the task-relevant object, limited influence from task-irrelevant high-saliency regions, and geometric consistency between the proposed motion and supported region.

7 Discussion

Has Existing Robustness Evaluation Already Addressed Visual Bias? Visual bias lies within robustness, but aggregate scores cannot distinguish failures caused by degraded observations from those caused by reliance on irrelevant evidence. Existing evaluations cover clutter, viewpoint, lighting, and other shifts [9, 34], while strong vision and vision-language models still exploit shortcuts and spurious correlations [5, 11, 44, 49]. Thus, robustness does not establish task-relevant grounding. Our controlled variants preserve task semantics, target visibility, and physical feasibility, yet high average success still coexists with substantial variation across task-equivalent conditions. Preference-aligned distractors cause more failures than matched random noise, and Action-Diff links detected content to action changes. These results motivate reporting visual bias separately as a safety primitive and expanding preference discovery and visual-shortcut search to improve audit coverage.

Can Findings from Manipulation Generalize to Broader Embodied AI?. Our manipulation experiments do not establish a universal bias profile, but the VLM-driven planning and grounding pipelines, end-to-end VLA policies, and related components considered here also support driving [43, 48, 59], navigation [36, 58], and diverse embodiments [47, 53]. Pretrained visual preferences may likewise influence physical decisions through these components, though we have not yet tested this in navigation or driving. Successful deployment with redundant sensing, runtime monitoring, human oversight, or downstream control does not imply that the underlying policy's visual bias has been eliminated. As foundation policies are reused across tasks and environments, independently auditing their visual-evidence dependence can improve native safety margins and reduce continual scenario-specific patching.

8 Conclusion

This work identifies visual bias as a distinct safety concern in embodied AI. Through controlled visual variations, naturalistic distractors, and action-relevance analysis, we show that task-equivalent visual changes and inherited visual preferences can alter action generation and induce physical task failures. These findings reveal visual dependencies that aggregate task-success metrics may fail to capture. We further introduce a counterfactual detection method that combines persistent region selection with action changes after masking, helping distinguish behaviorally relevant preference-aligned content from visually salient but harmless regions. Together, our results support treating visual bias as a safety primitive that should be independently measured and reported, while motivating broader preference discovery and visual-shortcut search for deployment-time assurance.

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A Action-Diff Score Ablations

We ablate individual score terms while keeping region selection and the remaining terms fixed, and recalibrate thresholds per suite at approximately 10% clean-query FPR. Between original and masked predictions, JS Mean measures Jensen–Shannon divergence, while Argmax Flip measures the fraction of dimensions whose most-probable tokens change. Expected Shift and Entropy Delta measure absolute changes in decoded expected actions and entropy. All four terms are averaged across action dimensions and eight prediction steps. Collapse captures abnormally low variation in the original translation predictions relative to clean references.

Table 2: Selected Action-Diff score ablations on OpenVLA ($K = 20$). Entries are episode-level TP/FP; Δ is relative to Full. Each suite contains 10 patched and 10 clean episodes; Long denotes LIBERO-10.

Variant	Long	Goal	Object	Spatial	Total	Δ
Full	10/4	4/1	10/3	10/2	34/10	–
w/o JS Mean	10/8	5/2	10/3	10/2	35/15	+1/+5
w/o Argmax Flip	10/7	5/2	10/3	10/2	35/14	+1/+4
w/o Expected Shift	10/4	5/2	10/3	10/2	35/11	+1/+1
w/o Entropy Delta	10/4	5/2	10/3	10/2	35/11	+1/+1
w/o Collapse	8/4	2/2	10/3	9/3	29/12	-5/+2
Region-Trigger	10/10	10/8	10/7	10/7	40/32	+6/+22

As shown in Table 2, removing Collapse worsens both detection and false-alarm control (34/10 to 29/12 TP/FP), whereas removing JS Mean or Argmax Flip mainly increases false positives (+5 and +4, respectively). Expected Shift and Entropy Delta have smaller effects at this operating point. Full yields 22 fewer false positives than Region-Trigger at the cost of six fewer true positives.